

ELEMENTS OF PROJECT MANAGEMENT

UNIT 1 + UNIT 2 — COMPLETE EXAM STUDY GUIDE

Actual questions • Bloom's levels • Notes-based answers • Diagrams • Memory tricks • Study plan

Prepared from the uploaded Unit 1 assignment/notes and Unit 2 Project Organization notes. Unit 2 previous-exam questions are reproduced from the question-paper pages in the uploaded Unit 2 PDF.

IMPORTANT

Bloom's levels shown for Unit 1 are the levels printed on the assignment. Unit 2 previous-exam questions include the BL levels visible in the uploaded question-paper images. Unit 2 "Additional Questions" do not have BL levels in the notes, so they are marked "BL — not specified" rather than inventing a professor-assigned level.

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1. BLOOM'S TAXONOMY — HOW TO WRITE FOR THE MARKS

BL	Meaning	What to do	Best answer shape
BL1	Remember	Recall definitions, names, lists, phases.	Definition → list
BL2	Understand	Explain meaning and significance.	Definition → explanation → simple example
BL3	Apply	Use the concept in a project/situation.	Concept → steps → application
BL4	Analyze	Compare, connect causes/effects, evaluate consequences.	Concept → factors → relationships → consequences → example → conclusion
Rule: BL1 is not an invitation to write a two-page essay. BL4 is not satisfied by a list. Match the answer depth to the verb and Bloom level.			

Fast writing templates

BL1: exact definition + required points.

BL2: meaning + why it matters + one example.

BL3: concept + procedure/components + apply to a real project.

BL4: define + break down + connect variables + consequences + example + conclusion.

2. UNIT 1 — ACTUAL ASSIGNMENT QUESTIONS + EXAM ANSWERS

The Unit 1 assignment is the five-page assignment issued by Dr. Rajesh Patil for AY 26–27. It explicitly labels each question with its Bloom level.

Q1. Write different definitions of project. [BL1]

- A project is a temporary endeavor undertaken to create a unique product, service, or result.
- A project is a set of temporary and interrelated tasks undertaken to produce a unique product, service or result.
- A project is a collection of linked activities carried out in an organized manner, with a defined start and end, to achieve specific results.
- Key words to memorize: temporary, unique, objective, start, end.

Q2. Write different definitions of project management. [BL1]

- Project management is the application of knowledge, skills and tools necessary to achieve project requirements.
- It is a controlled and structured process using organizational resources to achieve clearly defined objectives.
- Major PM processes: initiation, planning, execution, monitoring and control, and closure.

Q3. List the conditions which need to be satisfied to consider tasks/activities as a project. [BL3]

- Specific objective/business value.
- Defined specifications.
- Defined start and end dates.
- Funding/budget limits where applicable.
- Use of human and non-human resources such as people, money and equipment.
- Multifunctional nature — work may cross several functional departments.
- Application: a banking-app development effort has a defined objective, deadline, resources, budget and cross-functional team, so it qualifies as a project.

Q4. Explain benefits of studying project management for engineers OR explain why engineers irrespective of specialization must acquire PM knowledge/skills. [BL4]

- Planning and scheduling: engineers can organize activities, resources and deadlines.
- Scope management: engineers can understand boundaries and manage changes.
- Cost control: resources and costs can be estimated and monitored.
- Risk management: assumptions, risks and unexpected issues can be identified early.
- Resource optimization: people, equipment, money and materials can be coordinated.
- Communication: engineering projects involve multiple teams and stakeholders.
- Monitoring/control: actual progress can be compared with planned progress and corrective action taken.
- Conclusion: PM combines engineering knowledge with planning, coordination, resource management and decision-making.

Q5. Describe classification of projects. [BL1]

- Ownership: Public, Private, Public-Private Partnership.
- Investment: Large-scale, Medium-scale, Small-scale.
- Sector: Agricultural, Industrial, Service.
- Objectives: Commercial, Social.
- Nature: Conventional, Innovative.

- Time: Long-term, Medium-term, Short-term.
- Functions: Marketing, Financial, HR, IT & Technology, Production, Strategic.
- Risk: High-risk, Low-risk.
- Investment decisions: Independent, Mutually Exclusive, Contingent.
- Output: Quantifiable, Non-quantifiable.
- Other classifications in the notes include techno-economic characteristics and financial-institution categories.

Q6. Describe classification of project management in different environments. [BL1]

- Construction Project Management.
- Manufacturing Project Management.
- IT Project Management.
- Biotechnology Project Management.
- Localization Project Management.
- Memory: Construction → Manufacturing → IT → Biotechnology → Localization.

Q7. List and explain different Project Management approaches. [BL3]

- Traditional: plan, execute and control tasks to achieve goals within budget and time.
- Agile: collaborative, flexible and feedback-driven; emphasizes small batches, customer collaboration, quick feedback and transparency.
- Waterfall: sequential and linear; suitable when requirements/processes are well defined; difficult to modify later.
- Systems approach: views the project as an interconnected whole with feedback and interaction between subsystems.
- Critical Path/Critical Chain: critical path focuses on the longest activity sequence; critical chain additionally considers resource limitations.
- Program management: handles multiple projects, especially where functional improvement is the objective.

Q8. “Scope” and “Feasibility” are important concepts in project management. Explain in detail. [BL4]

- Scope defines what the project must accomplish: deliverables, tasks, deadlines, costs, inclusions and exclusions.
- Scope management includes planning, definition, verification and control.
- Feasibility evaluates whether a proposed project is viable and worth undertaking before major resources are committed.
- Core feasibility in the notes: technical, operational and economic; complete analysis also considers managerial, environmental, social/political and financial factors.
- Relationship: scope defines what is proposed; feasibility tests whether the proposal is realistically achievable.
- Example: solar plant scope defines capacity/site/deliverables/deadline; feasibility checks technology, manpower, operating capability, economics and environmental/funding conditions.

Q9. Elaborate on systems approach in project management with suitable example. [BL1]

- The systems approach treats the project as a whole rather than isolated parts.
- System elements include subsystems, environment, system boundary, input-output conversion and feedback.
- Project-management systems emphasize the project team and interacting, interdependent subsystems; their interaction affects efficiency and effectiveness.
- Example: in a metro project, engineering, procurement, finance, construction and stakeholder activities interact, so a change in one area can affect others.

Q10. List and explain phases of project life cycle. [BL3]

- Initiation: identify need, objectives, stakeholders and initial feasibility.

- Planning: define scope, activities, resources, cost, schedule and risks.
- Execution/Implementation with monitoring and control: perform the planned work while comparing actual performance with the plan and taking corrective action.
- Closure: complete/handover deliverables, close activities/contracts and document/evaluate the project.

Q11. Sketch and explain the common stretched S-project life cycle. [BL3]

- Draw an S-shaped progress-versus-time curve: slow start → rapid middle progress → flattening near completion.
- Start-up: project manager/team/resources are assembled and the work programme is organized.
- Middle: work accelerates and major activities are completed.
- End: progress slows as final tasks, integration and closeout activities take disproportionate effort.

Q11 (second item in assignment). Sketch and explain System Development Cycle. [BL3]

- Four phases: Conception → Definition → Execution → Operation.
- Conception: identify the system idea/need.
- Definition: establish requirements, objectives and system definition.
- Execution: develop/construct/implement the system.
- Operation: place the completed system into operation.

Q12. Describe Triple Constraints of Project Management. [BL2]

- Time: completion within the required schedule.
- Cost: completion within the approved budget.
- Scope: delivery of the agreed project scope.
- The notes identify Quality as an additional/fourth constraint.
- Important relationship: changing one constraint can affect the others.

Q13. “In Project Management Cost is a function of Performance, Time and Scope.” Explain with suitable example. [BL4]

- Scope increase can require more resources and therefore more cost.
- Higher performance/quality requirements may require better technology, skills or testing.
- Reducing time can require additional people, overtime, equipment or parallel work.
- Example: a banking application gains extra modules and higher security requirements while the deadline is shortened; more developers, security specialists and testing resources can increase cost.
- Conclusion: cost must be managed together with performance, time and scope.

Q14. Explain need for efficient and effective project management. [BL4]

- Control scope and manage change.
- Deliver on time and within budget.
- Maintain project-team focus.
- Collect user requirements.
- Define the critical path.
- Estimate resources, time and costs.
- Communicate progress, risks and changes.
- Explore assumptions and prepare for unexpected issues.
- Document knowledge gained.
- Poor PM can cause delays, cost increases, resource waste, quality dissatisfaction and reputational damage.

Q15. Explain benefits of project management. [BL3]

- Clear identification of responsibilities.
- Identification of time limits for scheduling.
- Trade-off analysis.
- Measurement of accomplishments against plans.
- Early identification of problems and corrective action.
- Improved estimating capability for future planning.
- Knowing when objectives cannot be met or will be exceeded.

Q16. List and explain five criteria Cleland and King have suggested for determining when to use project management. [BL4]

- Important source note: the uploaded Unit 1 assignment asks this question and cites the PM references, but the five Cleland-and-King criteria are not clearly reproduced in the supplied Unit 1 notes visible to us.
- Do not invent a five-point list and label it as professor notes.
- Exam action: obtain the exact slide/textbook passage and insert the five criteria. Then answer BL4 by explaining why each criterion indicates the need for project management.

Q17. List the processes of SDLC applied to IT projects and briefly explain each process. [BL3]

- Conception: identify the need/idea for the system.
- Definition: establish requirements, objectives and system characteristics.
- Execution: develop, construct or implement the system.
- Operation: put the completed system into operation.
- Flow: Conception → Definition → Execution → Operation.

Q18. Explain Waterfall, Incremental and Iterative development models associated with IT projects. [BL3]

- Waterfall: sequential/linear stages; works best when requirements are well defined; changes later are difficult.
- Incremental: deliver the final system in multiple increments, each adding usable functionality.
- Iterative: develop through repeated cycles, refining/improving the solution through feedback.
- Key distinction: incremental adds functionality in pieces; iterative improves/refines through repeated cycles.

Q19. Explain six-constraint model: Cost, Risk, Benefits, Quality, Scope and Time. [BL3]

- Cost — stay within financial limits.
- Risk — identify/manage uncertainty that can affect the project.
- Benefits — achieve expected stakeholder/organizational benefits.
- Quality — satisfy required performance/quality standards.
- Scope — deliver agreed products/services/results.
- Time — complete within the required schedule.
- Relationships: scope can increase cost/time; reduced time can increase cost/risk; reduced cost can threaten quality; risk can threaten time, cost and benefits.

Q20. Explain quality in project management, particularly focusing on IT projects. [BL4]

- Quality means deliverables meet requirements, specifications and expected performance.
- IT quality dimensions to discuss: requirements, functionality, performance, reliability, security, testing and maintainability.
- Quality interacts with time, cost and scope.

- Example: rushing deployment by reducing testing may save time initially but can increase defects and reduce reliability.
- Conclusion: quality must be balanced with other project constraints.

Q21. Apply Project Life Cycle to an engineering project. [BL3]

- Use solar power plant as the example.
- Initiation: identify need, objective, stakeholders and initial feasibility.
- Planning: define scope, site, budget, schedule, resources, procurement and risks.
- Execution + monitoring/control: site preparation, procurement, installation, electrical work and testing; monitor time, cost, scope, quality and progress.
- Closure: final testing, handover, documentation, contract closure and evaluation.

Q22. Analyze significance of each phase of Project Life Cycle. [BL4]

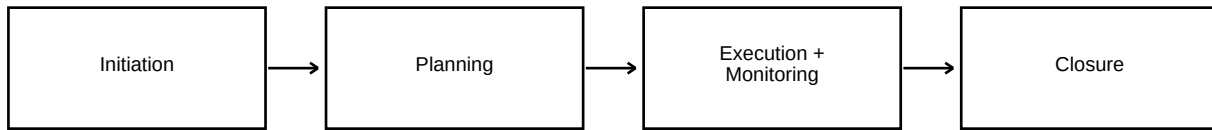
- Initiation gives direction; poor objectives can send the project in the wrong direction.
- Planning defines scope/resources/cost/schedule/risks; poor planning can cause shortages, delays and overruns.
- Execution/monitoring converts plans into deliverables and enables corrective action; poor execution can cause defects and waste.
- Closure ensures handover, documentation, contract closure and evaluation.
- Overall: phases are interdependent, so weakness in one phase can propagate into later performance.

Q23. Analyze importance of a feasibility study before initiating an engineering project. [BL4]

- Technical feasibility: technology, equipment and manpower.
- Economic feasibility: costs versus expected benefits.
- Operational feasibility: ability to operate and maintain the system.
- Managerial/administrative feasibility: organizational capability.
- Environmental feasibility: environmental conditions/impact.
- Social/political feasibility: demographic, social and political context.
- Financial feasibility: funding needs and sources.
- Decision: these factors collectively support approval, modification or rejection before major resources are committed.

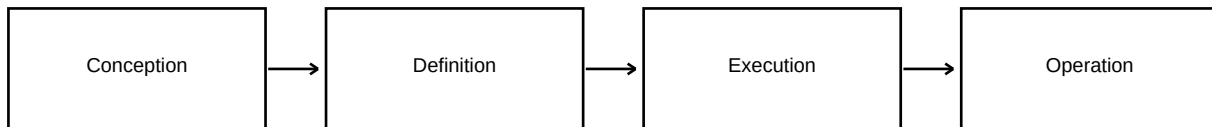
UNIT 1 — DIAGRAMS YOU SHOULD BE ABLE TO DRAW

A. Project Life Cycle



Draw the phases left-to-right. Under each box, write 2–3 keywords.

B. System Development Cycle



C. Triple Constraint triangle

Draw a triangle with **Time**, **Cost**, and **Scope** at the three corners. Write **Quality** inside/alongside as the additional constraint identified in the notes.

D. Stretched S-curve

Sketch: low initial slope → steep middle slope → flattening near completion. Label axes: Progress (Y) and Time (X).

E. Systems approach

Draw: Project Team in the centre, connected to interacting subsystems; show Environment around the system and arrows for inputs, outputs and feedback.

UNIT 1 MEMORY SHEET

- Project = Temporary + Unique + Objective + Start/End.
- PM processes = Initiation → Planning → Execution → Monitoring/Control → Closure.
- Life Cycle = Initiation → Planning → Execution/Monitoring → Closure.
- System Development = Conception → Definition → Execution → Operation.
- Triple = Time + Cost + Scope; Quality is an additional constraint.
- Feasibility core = Technical + Operational + Economic; complete analysis adds Managerial, Environmental, Social/Political and Financial.
- BL4 formula = Relationship + consequence + example + conclusion.

3. UNIT 2 — PREVIOUS EXAM QUESTIONS + NOTES-BASED ANSWERS

The Unit 2 PDF contains previous MPSTME exam-question pages. The Bloom levels below are taken from the visible question-paper images, not guessed.

1. Compare and contrast the functional, matrix, and projectized project organization structures. [BL2]

- Functional/traditional: organized around functions such as engineering, manufacturing, marketing, HR and IT. Employees normally report through the functional hierarchy.
- Projectized/pure product: project manager has final authority; team members work full-time on the project and report directly to the project manager.
- Matrix: combines functional and projectized structures; team members can have two bosses — functional manager and project manager. Strong matrix gives more power to PM; weak matrix gives more power to functional manager.
- Comparison: functional gives strong functional expertise/control; projectized gives strong project authority/focus; matrix balances project focus with functional resource sharing but creates dual-reporting complexity.

2. Define the roles and responsibilities of project managers and project teams. How do they contribute to project success? [BL1]

- Project manager: organizes, plans and executes projects within constraints such as budget and schedule; manages people, tasks, schedules and resources; leads the team; defines goals; communicates with stakeholders; integrates everything/everybody to achieve project goals.
- Responsibilities: plan tasks/end results, create WBS/schedule/budget, allocate resources, select/organize team, negotiate with stakeholders, monitor status, identify problems, solve problems and resolve conflicts.
- Project team: a small group with complementary skills committed to a common purpose, performance, goals and approach, accountable directly or indirectly to the PM.
- Success contribution: PM integrates people/resources; team contributes complementary skills, commitment, collaboration and execution.

3. Explain the roles and responsibilities of a project manager in a project organization. [BL2]

- Plan project tasks and end results.
- Create WBS, schedule and budget.
- Coordinate tasks and allocate resources.
- Select and organize the project team.
- Work and negotiate with stakeholders and manage expectations.
- Monitor project status and communicate status.
- Identify technical/functional problems.
- Solve problems or find appropriate support.
- Deal with crises and resolve conflicts.
- Prime role: integrate everything and everybody to accomplish project goals.

4. Differentiate between various types of project organization structures. Provide examples of when each structure is most suitable. [BL2]

- Functional/traditional: best suited to smaller organizations with closely located sections and a small number of goods/services. Example: a small manufacturing company organized into production, marketing, HR and IT.
- Projectized: suited when project focus and PM authority are critical. Example: a major product-development project with a dedicated full-time team.

- Matrix: suited to large/complex work where people from multiple functions must be shared across projects. Example: a technology company running several projects using shared engineering, marketing and finance specialists.
- Exam comparison: functional = functional authority; projectized = PM authority; matrix = shared authority.

5. Explain the role of the project manager and his team in detail. [BL2]

- PM leads, integrates, plans, coordinates, allocates resources, communicates, monitors and resolves problems/conflicts.
- Team provides complementary skills and is committed to common purpose, performance, goals and approach.
- Team-building is especially important when backgrounds/responsibilities are varied or when projects involve innovation, high risk and tight schedules.
- Effective teams improve communication, collaboration, creativity, trust and role clarity.

6. Explain the role of project manager in detail and discuss interpersonal skills the project manager must possess. [BL2]

- Role: plan, organize, allocate resources, coordinate, negotiate, monitor, communicate, solve problems and resolve conflicts.
- Interpersonal/leadership dimensions in the notes include communication, team building, leadership, conflict resolution, emotional intelligence, stakeholder management and collaboration.
- Effective interpersonal skills help the PM integrate people across functional lines and maintain cooperation.

7. Discuss the interpersonal skills required to be a successful project manager and describe how each contributes to success. [BL2]

- Team building: creates commitment, team spirit and collaboration.
- Leadership: integrates individuals into decisions and guides the team in an unstructured environment.
- Conflict resolution: reduces detrimental conflict and improves cooperation.
- Communication: keeps goals, decisions, progress and problems understood across levels.
- Emotional intelligence: helps regulate emotions, understand others and improve communication/conflict resolution.
- Stakeholder interaction: supports expectation management and project alignment.

8. Describe the role of structure of the organization in project management. [BL2]

- Organizational structure defines roles, responsibilities, authority relationships and communication channels.
- It establishes reporting relationships and the official chain of control.
- It affects how work is assigned, coordinated and controlled.
- A well-designed structure supports teamwork, transparency, accountability and smooth workflow.
- Poor structure can create confusion, lower productivity, burnout and poor outcomes.
- Structure selection affects PM authority, resource assignment, communication and project-team independence.

9. Explain types of project managers and their roles. [BL2]

- Prophet: pursues opportunities outside existing strategic boundaries without trustworthy quantitative evidence; gains followers around a grand vision.
- Gambler: pursues opportunities within strategic boundaries without a strong evidence-based business case.
- Expert: pursues opportunities outside strategic boundaries where trustworthy data supports a solid business case.
- Executor: pursues opportunities within existing strategy with strong business cases and trustworthy quantitative evidence.
- Memory: Prophet = outside/no data; Gambler = inside/no data; Expert = outside/data; Executor = inside/data.

10. Describe the parameters on which organization structure depends. [BL2]

- Rate of change of technology.
- Availability of resources.
- Product/service sold.

- Competition.
- Decision-making requirements.
- Size of organization.
- Environment: legal, political, social and economic.
- Culture: shared values, beliefs, customs and practices.

11. Describe qualifications/skills set required for project manager and its impact on success/failure of project. [BL4]

- Core skill set: team building, leadership, conflict resolution, technical expertise, planning, organization, entrepreneurship, administration, management support and resource allocation.
- Team building creates commitment, trust and collaboration.
- Leadership integrates people and supports decision-making.
- Conflict resolution prevents detrimental disputes from causing delays and poor decisions.
- Technical expertise helps evaluate solutions and trade-offs among cost, schedule and technical issues.
- Planning/organization create the roadmap, reporting structure and resource coordination.
- Administrative/support/resource skills enable budgets, schedules, meetings, reports, management relationships and resource commitments.
- Therefore, weaknesses in these skills can propagate into coordination problems, conflict, delays and poor project performance.

12. Describe pure product (projectized) organization. [BL2]

- Project-focused structure where the PM has final authority over project decisions, priorities and resources.
- PM leads the group and has ultimate authority for project decisions.
- All work is viewed as a project; team members report directly to PM.
- PM works with central authority/autonomy and the team is full-time on the project.
- Advantages: strong authority, rapid response, strong communication, customer focal point and loyalty.
- Disadvantages: duplication of effort/cost, possible retention of people beyond need, weaker technology continuity, limited technical interchange and career continuity.

13. Explain selection of project manager as a critical task for top management and list screening parameters. [BL4]

- Selection is difficult because the PM integrates people, resources, stakeholders and project constraints.
- Top management generally establishes a screening committee, selection criteria and policy, with senior management involvement.
- Screening parameters: background and experience, leadership and strategic expertise, technical expertise, people-management skills.
- Personal characteristics emphasized in the notes include flexibility, initiative, confidence, communication/integration ability, ability to balance technical solutions with time/cost/human factors, organization, problem identification and decision-making.
- Analysis: a technically strong candidate may still fail if unable to lead, integrate and make decisions under project constraints.

14. Describe advantages of matrix organization structure when one studies project management. [BL2]

- PM can maintain strong project control through line managers over resources, cost and personnel.
- Policies/procedures can be tailored to projects within company rules.
- Rapid response to changes and conflicts is possible.
- Functional organizations support the project.
- People retain a home after project completion and can have a career path.

- Shared key people minimize cost and allow specialists to work on varied problems.
- Strong technical base and knowledge can benefit multiple projects.
- Authority/responsibility are shared and there can be better balance among time, cost and performance.

15. Define Project Manager and analyze importance/significance of emotional intelligence on project success and outcomes. [BL4]

- PM is the professional who organizes, plans and executes projects within constraints and integrates everything/everybody to achieve project goals.
- Emotional intelligence (EI) is the ability to monitor own and others' feelings/emotions, distinguish between them and use that information to guide thought and action.
- EI includes evaluation/expression of emotions, regulation and use of emotional information in conflict resolution.
- Goleman's five components in the notes: self-awareness, self-regulation, motivation, empathy and social skills.
- Impact: stronger communication, conflict resolution, stakeholder relationships and team cohesion can improve project outcomes.
- Conclusion: effective PM is not purely technical; emotional/interpersonal capability contributes directly to project and team success.

16. Compare traditional (classical) organization structure with projectized (pure product) organization structure. [BL4]

- Authority: functional manager vs PM.
- Team allocation: functional employees may work across projects vs full-time project team.
- Reporting: vertical functional chain vs direct reporting to PM.
- Project focus: lower project orientation vs strong project orientation.
- Response: functional decisions can require longer coordination vs rapid project response.
- Technical continuity: strong functional groups vs risk of weaker cross-project technical interchange.
- Cost: functional sharing can be efficient vs projectized duplication can be expensive in multiproduct organizations.
- Use a two-column table in the exam.

4. UNIT 2 — ADDITIONAL QUESTIONS FROM THE NOTES

These ten questions appear under “Additional Questions” in the Unit 2 PDF. No Bloom level is printed for them, so they are marked BL — not specified.

A1. “Selection of project manager is critical, important, challenging task for top management.” Explain. [BL — not specified in source]

- PM selection is difficult because the PM integrates project people, tasks, resources, stakeholders and constraints.
- Top management normally establishes a committee, selection criteria and policy, and involves senior management.
- Candidate screening includes background/experience, leadership/strategic expertise, technical expertise and people-management skills.
- Personal characteristics are heavily emphasized: adaptability, initiative, confidence, communication, integration, organization, decision-making and balancing technical/time/cost/human factors.
- Conclusion: the wrong PM can create leadership, coordination and decision problems across the whole project.

A2. Describe the selection process of Project Manager. [BL — not specified in source]

- Establish screening/selection committee.
- Set professional qualifications and selection criteria.
- Follow top-management selection policy.
- Involve senior management.
- Screen candidates using background/experience, leadership/strategic expertise, technical expertise and people-management skills.
- Assess personal characteristics such as flexibility, communication, initiative, decision-making and balance of technical/time/cost/human factors.

A3. List and explain expectations from core team members of a project. [BL — not specified in source]

- Commitment to the project.
- Sharing responsibility.
- Adaptability.
- Task orientation.
- Staying on schedule.
- Trustworthiness and supportiveness.
- Team orientation.
- Open-mindedness.
- Flexibility across structure and authority.
- Use of project-management tools.

A4. List requirements the PM should consider while selecting contracted team members. [BL — not specified in source]

- Determine required skill set and number of personnel.
- Identify companies/contractors/manpower consultancies to invite.
- Set proposal-analysis and selection criteria.
- Choose the contractor/consultant.
- Enter into the contract after selection.

A5. Discuss problems that may arise due to contracted team members. [BL — not specified in source]

- Short-term work must be carefully integrated with the project schedule.
- Roles and relationships with other activities must be clearly defined.
- Commitment may be lower and quality standards may be violated.
- Greater supervision/monitoring may be required, adding responsibility for the PM.

A6. Write a detailed note on project office. [BL — not specified in source]

- Project office can mean a support staff group reporting to the PM and the physical place where the team meets.
- Functions/composition depend on PM authority and project size, importance and goals.
- Functions: information centre for internal/external stakeholders; control time, cost and performance against contract; document and communicate tasks; ensure authorization/funding through contract documents; integrate work across functional departments.
- When one office coordinates multiple projects and has larger staff, it is called a Project Management Office (PMO).

A7. Explain why integration in large-scale/mega projects is important and significant. [BL — not specified in source]

- Mega projects are large, complex, multi-year ventures involving many stakeholders.
- Integration identifies, defines, combines, unifies and coordinates project processes and activities.
- The difficult part is coordinating project elements so joint scope, schedule and budget goals are met.
- Many parties contribute: sponsors, prime contractors, subcontractors, consultants and suppliers.
- Success dimensions in the notes: efficiency, customer/user impact, business/financial success, and societal impact.
- Therefore, poor integration can create interface problems and undermine the total project even if individual components perform well.

A8. Explain importance and significance of emotional intelligence of PM on project success. [BL — not specified in source]

- EI includes monitoring and distinguishing own/others' emotions and using emotional information to guide action.
- It supports emotional regulation and conflict resolution.
- Goleman's five components: self-awareness, self-regulation, motivation, empathy, social skills.
- High EI can improve communication, conflict resolution, stakeholder relationships and team cohesion.
- Conclusion: PM effectiveness requires both technical and interpersonal/emotional capability.

A9. Why is PM experience more important/valued for big/mega projects than educational qualification? [BL — not specified in source]

- Mega projects involve high complexity, uncertainty, stakeholder conflict and high stakes.
- Experience provides real-world problem solving, leadership under pressure, stakeholder management, risk/change management and team/resource coordination.
- Top management values a track record of successful delivery.
- Education provides theoretical foundations, but the notes state it does not alone provide practical wisdom for real-time decisions, human dynamics and political challenges.
- Therefore, for large/mega projects, experience becomes especially valuable while education remains foundational.

A10. Explain how project team affects success as well as failure of a project. [BL — not specified in source]

- A project team is a small group with complementary skills committed to common purpose, goals and approach.
- Good team building improves communication, conflict resolution, collaboration, creativity, trust and role clarity.

- Team members are expected to be committed, adaptable, responsible, trustworthy, open-minded and schedule-oriented.
- Tuckman stages: Forming → Storming → Norming → Performing → Adjourning.
- Poor team dynamics can create communication breakdown, role misunderstandings, morale issues, conflict and delays.
- Strong team dynamics move the group toward Performing and improve project success.

5. UNIT 2 — DIAGRAMS YOU SHOULD DRAW

A. Functional / Traditional Structure

Draw a top box such as CEO/Director, then vertical functional boxes: Marketing | Production | Human Resources | Information Technology | Customer Service.

Use when explaining the functional structure. Label the vertical reporting lines.

B. Projectized / Pure Product Structure

CEO / Chief Executive ↓ Project Manager → Staff ↓ Project Manager → Staff ↓ Project Manager → Staff

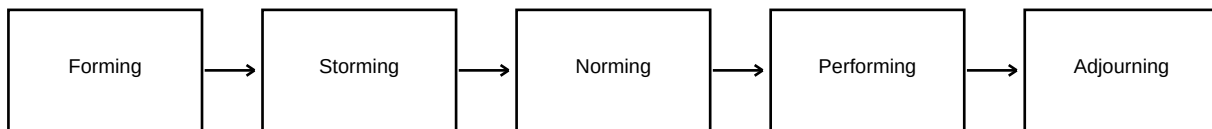
Main label: PM has final authority; team members report directly to PM and work full-time on the project.

C. Matrix Structure

General Manager / CEO ↓ Functional Managers: Marketing | R&D | Engineering | Production ↑ Project Managers: Project 1 | Project 2 | Project 3 ↓ Functional Staff assigned to projects

Highlight the key idea: dual reporting — functional manager + project manager. Strong matrix = more PM power; weak matrix = more functional-manager power.

D. Tuckman Team Development Model



This is a very high-value diagram for team-building questions.

E. Project Office

Project Manager ↓ Project Office / Support Staff ↑ Functional Employees / Departments ↓ Internal + External Stakeholders

Write the functions beside it: information centre; time/cost/performance control; documentation; authorization/funding; functional integration.

F. Mega-Project Integration

Sponsors ↔ Prime Contractors ↔ Subcontractors ↔ Consultants ↔ Suppliers ↓ ONE PROJECT GOAL ↓ Scope + Schedule + Budget + Outcomes

Add the four success dimensions: Efficiency, Customer/User Impact, Business/Financial Success, Societal Impact.

6. HIGH-YIELD COMPARISON TABLES

Feature	Functional / Traditional	Projectized	Matrix
Authority	Functional manager	Project manager	Shared
Reporting	Mainly functional/vertical	Directly to PM	Two bosses possible
Team	Functional pool; may support projects	Full-time project team	Shared functional resources
Project focus	Lower	Very high	High, but balanced with functions
Communication	Vertical	Strong project channels	Horizontal + vertical
Response	Can be slower due to functional approvals	Very rapid	Can be rapid but complex
Main strength	Technical specialization / resource sharing	Strong project control	Balance + shared specialists
Main weakness	No single project authority	Duplication/cost and weaker technical interchange	Dual reporting/conflict/role ambiguity

Project Manager vs Project Team

PM = integrates, plans, allocates, communicates, negotiates, monitors, solves and leads. Team = contributes complementary expertise, executes work, shares responsibility and collaborates toward common goals.

Core team vs Contracted team

Core team = functional/internal members expected to commit, share responsibility, adapt, stay on schedule and use PM tools. Contracted team = external personnel hired when internal staff/expertise is insufficient; they require careful integration, clear roles and more monitoring.

EI vs Technical Expertise

Technical expertise supports evaluation of technical concepts, communication in technical terms and cost/schedule/technical trade-offs. EI supports emotion regulation, empathy, communication, conflict resolution and stakeholder/team relationships.

7. MEMORY TRICKS — UNIT 1 + UNIT 2

- UNIT 1 Project = T-U-O-S/E: Temporary, Unique, Objective, Start/End.
- PM Life Cycle = I-P-E-C: Initiation, Planning, Execution, Closure.
- System Development = C-D-E-O: Conception, Definition, Execution, Operation.
- Triple constraints = T-C-S: Time, Cost, Scope. Add Q for Quality.
- Feasibility core = T-O-E: Technical, Operational, Economic.
- Functional vs Projectized vs Matrix = F-P-M: Function controls / Project manager controls / Mixed shared control.
- Matrix = Two bosses: Functional + Project.
- Projectized = PM has power + team is full-time.
- Tuckman = F-S-N-P-A: Forming, Storming, Norming, Performing, Adjourning.
- PM skill set = T-L-C-T-P-O-E-A-M-R: Team building, Leadership, Conflict resolution, Technical, Planning, Organization, Entrepreneurial, Administration, Management support, Resource allocation.
- PM types = P-G-E-E: Prophet, Gambler, Expert, Executor. Think: outside/inside × evidence/no evidence.
- Four mega-project success dimensions = E-C-B-S: Efficiency, Customer/User impact, Business/Financial success, Society.

8. 7-DAY STUDY PLAN

Day	Focus	What to do
Day 1	Unit 1 definitions + classification	Learn Q1, Q2, Q5, Q6. Active recall twice.
Day 2	Unit 1 concepts	Q7–Q10 + scope/feasibility/systems approach. Draw 3 diagrams.
Day 3	Unit 1 application/analysis	Q11–Q23. Focus on BL3/BL4. Write Q13, Q22, Q23 once.
Day 4	Unit 2 organization structures	Functional, Projectized, Matrix. Draw and compare all three.
Day 5	Unit 2 PM + team	PM roles, skill set, EI, conflict, core/contracted team, Tuckman.
Day 6	Unit 2 advanced	Project office, mega-project integration, selection of PM, experience, PM types.
Day 7	Mock exam	No notes. Answer 2 BL1, 2 BL2, 2 BL3, 2 BL4 questions + diagrams. Review mistakes.

Daily study method — 60/30/10

- 60 minutes: learn new material.
- 30 minutes: active recall without notes.
- 10 minutes: draw diagrams from memory.
- Do not repeatedly reread. Recall → check → correct → recall again.

9. LAST-NIGHT / 6-HOUR REVISION PLAN

Time	Task
Hour 1	Definitions + lists: Q1/Q2/Q5/Q6; Unit 2 structures.
Hour 2	Project Life Cycle, System Development Cycle, approaches, feasibility, constraints.
Hour 3	Functional vs Projectized vs Matrix + PM role/skills.
Hour 4	Team building, Tuckman, conflict, EI, project office, contracted/core team.
Hour 5	Mega-project integration + Unit 1 BL4 questions + Unit 2 BL4 questions.
Hour 6	Diagrams + active recall + write one full BL4 answer.

If you only have 2 hours

1) Memorize all flow diagrams. 2) Memorize Functional/Projectized/Matrix comparison. 3) Learn PM roles + 10 skills. 4) Learn Tuckman. 5) Learn feasibility + six constraints. 6) Practice one BL4 answer.

10. ACTIVE-RECALL SELF-TEST

1. Define project in three ways.
2. Define project management.
3. List conditions that make an activity a project.
4. Classify projects.
5. Name PM approaches.
6. Explain scope vs feasibility.
7. Draw Project Life Cycle.
8. Draw System Development Cycle.
9. Explain triple constraints.
10. Explain cost as a function of performance, time and scope.
11. List PM skill set from Unit 2.
12. Differentiate functional, projectized and matrix structures.
13. Why does matrix have dual reporting?
14. Define project team.
15. List expectations from core team members.
16. List requirements for selecting contracted team members.
17. List problems with contracted team members.
18. Write functions of project office.
19. Draw Tuckman model.
20. Explain four types of PM.
21. Define EI and list Goleman's five components.
22. Explain why experience matters in mega projects.
23. Define mega project and list four success dimensions.
24. Explain integration in large-scale projects.

11. FINAL EXAM CHECKLIST

- ■ I can write project and project-management definitions without looking.
- ■ I know the Unit 1 assignment Bloom level for every question.
- ■ I can explain scope and feasibility and their relationship.
- ■ I can draw the Project Life Cycle and System Development Cycle.
- ■ I can explain the triple constraints and six-constraint model.
- ■ I can compare Functional, Projectized and Matrix structures in a table.
- ■ I know the project manager's responsibilities.
- ■ I know all 10 PM skills listed in Unit 2.
- ■ I know the expectations from core team members.
- ■ I know contracted-team selection criteria and problems.
- ■ I can explain project office/PMO.
- ■ I can draw Tuckman's five stages.
- ■ I can explain EI and its five components.
- ■ I can explain mega-project integration and four success dimensions.
- ■ I can answer at least one BL4 question using cause → effect → consequence → example.
- ■ I have checked Q16 against the exact professor/textbook source because its five Cleland-and-King criteria are not reproduced in the supplied notes.

SOURCE / SCOPE NOTE

Unit 1: based on the uploaded "Assignment One-Unit One July26 Nov26 for students.pdf" and the uploaded Unit 1 PM notes. Unit 2: based on the uploaded "Unit 2 Revised January 26.pdf" / duplicate copy, including the previous-exam question pages and the additional-question page. Where a Unit 2 question has no printed BL in the source, this guide does not fabricate one.